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MILITARY SPECIFICATION

RELAY, SENSITIVE

(FOR TORPEDO MARK 37 MOD 2)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 Scope. This specification describes the requirements for the procurement of a double-pole, double-throw, sensitive electromagnetic relay, Drawing 1521140. The coil is rated at one watt at a voltage of 31.6 volts dc. The relay is used in the control of functional circuits which operate on a nominal dc voltage of 25 volts and dc current up to two amperes.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

O-S-642	Sodium Phosphate, Tribasic, Technical
FF-N-836	Nuts, Hexagon and Square
FF-S-86	Screw Cap, Socket-Head
FF-S-92	Screws, Machine, Slotted or Cross-Recessed
QQ-C-320	Chromium Plating (Electrodeposited)
QQ-N-290	Nickel Plating (Electrodeposited)

FSC 5945

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QQ-S-365	Silver-Plating (Electrodeposited)
QQ-S-571	Solder, Soft (Tin, Tin-Lead, Lead-Silver)
QQ-Z-325	Zinc Plating (Electrodeposited)
TT-C-490	Cleaning Methods and Pretreatment of Ferrous Surfaces For Organic Coatings
TT-I-558	Ink, Stencil, Opaque, for Marking Non-Porous Surfaces
TT-L-58	Lacquer; Spraying, Clear and Pigmented
TT-P-664	Primer, Synthetic, Lacquer-Resisting
TT-T-291	Thinner, Paint, Volatile Mineral Spirits
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner

Military

MIL-P-116	Preservation, Methods of
MIL-B-857	Bolts, Nuts, and Studs
MIL-M-3171	Magnesium Alloy, Processes for Corrosion Protection of
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-S-6872	Soldering Process, General Specification
MIL-B-7883	Brazing of Steel, Copper, Copper Alloys and Nickel Alloys
MIL-P-8585	Primer Coating, Zinc-Chromate, Low Moisture Sensitivity

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MIL-A-8625	Anodic Coating, for Aluminum and Aluminum Alloys
MIL-E-15090	Enamel, Equipment, Light-Gray (Formula 111)
MIL-C-15328	Primer (Wash), Pretreatment, Blue, (Formula 117-B for Metals)
MIL-V-16399	Varnish, Moistureproof
MIL-T-18404	Torpedoes, Environmental Requirements; General Specification for
MIL-E-82590	Environmental Requirements and Tests (For Torpedo Mk 37 Components and Assemblies)

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads, 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base

Naval Ordnance Systems Command

OSTD 600	Standard Parts

DRAWINGS

Naval Ordnance Systems Command

Drawing 1521140	Relay, Sensitive

PUBLICATIONS

Naval Ordnance Systems Command

OS 6341	Ordnance Design Requirements, General Miscellaneous

(Copies of specifications, standards, drawings, and publications requested by contractors in connection with specific procurement functions

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should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards Publications

Handbook H28	Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 Description. The sensitive relay (hereinafter called the relay), Drawing 1521140, is a sensitive, electromagnetic relay with a double-pole, double-throw armature. With the coil in its normally deenergized state, the armature closes the circuit(s) through one pair of poles. When the coil is energized and the coil current reaches a predetermined value, the armature moves to close the circuit(s) through the opposite pair of poles, at the same time opening the circuit(s) through the initial pair of poles. When the current in an energized coil is reduced below a predetermined value, the armature will return to the position assumed when the coil is deenergized. The coil is energized from a nominal 10 volts dc (or rectified ac) power source.

3.2 Definitions. Definitions applicable to the environmental requirements of this specification are contained in MIL-T-18404.

3.3 General requirements

3.3.1 Conflicting requirements. Conflicting requirements arising between this specification or any specifications, publications, or drawings listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation, clarification, and resolution (or correction).

3.3.2 Data required. No data is required by this specification (other than reports accompanying samples submitted for preproduction and periodic production testing), or by applicable documents referenced in Section 2, unless specified in the contract or order (see 6.2).

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3.3.3 Preproduction sample. Unless otherwise specified in the contract or order, preproduction samples of the relays shall be manufactured using the methods and procedures proposed for the production lot. These units shall be tested as specified in Section 4 herein and are for the purpose of determining, prior to starting production, that the contractor's production methods are capable of producing relays that comply with the technical requirements of the contract. See 4.3.1

3.3.4 Materials. Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering into the final assembly of the relay shall conform strictly to the applicable Federal or Military specifications. Materials which are not covered by specification or which are not specifically designated herein shall be of the best quality, or the lightest practicable weight consistent with strength and durability, and entirely suited to the purpose for which intended under all conditions of test and service operation. Recommendations concerning the use of a superior substitute for any part or material specified herein shall be accompanied by a description of the proposed substitute, the reasons therefore, together with evidence to substantiate any claim as to its suitability. At the discretion of the Naval Ordnance Systems Command, test samples may be required to prove the suitability of the proposed substitute.

3.3.4.1 Critical materials. Materials shall be selected on the basis of suitability and availability, taking into consideration the additional restrictions created during a time of national emergency. Critical materials shall be used only when essential for satisfactory operation of the particular item or equipment. Substitution of less critical materials for materials required by this specification is subject to approval by the Naval Ordnance Systems Command.

3.3.4.2 Magnesium. The use of magnesium requires the specific approval of the Naval Ordnance Systems Command or its appointed agency.

3.3.4.3 Flammable materials. Materials that support combustion or which are capable of causing an explosion shall not be used.

3.3.4.4 Toxic materials. Materials that produce harmful toxic effects under conditions encountered in the Naval Service shall not be used.

3.3.5 Moisture, fungus, and corrosion. The relay shall be rendered moisture, fungus, and corrosion resistant as required by NAVORD OS 6341.

3.3.6 Finishes and coatings

3.3.6.1 General. All machining operations shall be completed before the application of the corrosion-resistant treatment.

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3.3.6.2 Plating. Electrodeposited metallic coatings shall conform to the following. The plating thickness in all cases shall be adequate to ensure conformance with the parts or equipment specification for conductivity and corrosion resistance:

Zinc	QQ-Z-325, Type II, Class 2 or 3
Nickel	QQ-N-290
Chromium	QQ-C-320, Type I
Silver	QQ-S-365

NOTE

Cadmium plating shall not be used.

3.3.6.3 Tin and lead-tin alloys. Platings of tin and lead-tin alloys, when used, shall be of adequate thickness and so continuous as to pass the marine environment test.

3.3.6.4 Corrosion resistant steel

3.3.6.4.1 Cleaning. Remove all oil, grease, dirt and other foreign matter from the metal by vapor degreasing, washing with petroleum spirits, TT-T-291, Grade 1, or by immersion in a boiling solution of trisodium phosphate, O-S-642, followed by rinsing in cold water and a final rinse in hot water. If necessary because of scale, immerse in the following solution held at 60 degrees centigrade (C) (140 degrees Fahrenheit (F)) until the scale is removed:

Nitric acid	20 percent by volume
Hydrofluoric acid	1 to 3 percent by volume
Water	77 to 79 percent by volume

Then rinse in cold water and blow dry with compressed air. If structures are fabricated with overlapping joints the preferred method of cleaning is by vapor degreasing to avoid entrapment of caustic or acid solutions.

3.3.6.4.2 Passivation. Immerse the clean metal in a solution of equal parts of nitric acid (40 degrees Baume) and water, held at a temperature of 71 degrees C (160 degrees F), for five minutes. Rinse first in clean cold water followed by a rinse in hot water. Then blow dry with compressed air.

3.3.6.5 Anodic film. Parts fabricated from aluminum or aluminum alloys shall be anodized in accordance with MIL-A-8625 or chemically treated in accordance with MIL-C-5541 when surfaces are to be painted.

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3.3.6.6 Magnesium and magnesium alloys. Magnesium and magnesium alloys, when authorized, shall be finished in accordance with MIL-M-3171.

3.3.6.7 Painting. If painting is selected as the final finish, the following processes shall be used:

3.3.6.7.1 Cleaning. Surfaces to be painted shall be free of dirt, dust, grease, mill scale, welding flux, and other foreign matter. Cleaning may be accomplished with volatile grease solvent, vapor degreasing detergents, hot alkali cleaning solutions, or other processes which have no deleterious effect on the surfaces. Welding irregularities shall be removed. Alkali cleaning shall not be used on aluminum or aluminum alloy surfaces. When an alkali cleaning solution is used, the final rinse shall consist of a dilute chromic acid solution (0.5 pound of chromic acid per 100 gallons of water). When fabrication involves overlapping joints, vapor degreasing is the preferred method of cleaning.

3.3.6.7.2 Pretreatment. Steel surfaces to be painted shall be pretreated by either of the following:

(a) One coat of wash pretreatment in accordance with MIL-C-15328, applied as a continuous film 0.0004 to 0.0005 inch thick; or,

(b) A phosphate coating conforming to TT-C-490, Grade I.

3.3.6.7.3 Primer. All painted surfaces shall have a primer coat applied as a continuous film 0.0004 to 0.001 inch thick. For steel surfaces the primer shall be in accordance with TT-P-664 or MIL-P-8585, and for nonferrous surfaces the primer shall be in accordance with MIL-P-8585.

3.3.6.7.4 Enamel. Two coats of enamel, light gray, MIL-E-15090, Class 2, applied as a continuous film approximately 0.001 inch thick.

3.3.6.7.5 Exceptions. The following surfaces shall not be painted: mounting surfaces, pins, and threaded parts.

3.3.7 Marking. Marking shall be in accordance with Drawing 1521140. Marking shall be by means of engraving, metal stamping, embossing, by permanently affixed nameplate, or by marking with ink, stencil, TT-I-558. Ink shall be black or a color contrasting to the surface to which applied. Ink TT-I-558, when applied to an enamel surface, shall be covered with varnish, moistureproof, MIL-V-16399; when applied to other surfaces it shall be covered with lacquer, TT-L-58, Type I.

3.3.7.1 Marking data. Unless otherwise specified in the contract or order, or detailed in the applicable drawings, marking shall include the following:

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(a) In Area B, Drawing 1521140: name of item or assembly; drawing number and drawing revision letter; operating voltage or current; contact rating; coil resistance; manufacturer's name, symbol, or trade mark; and manufacturer's part or assembly number (if desired).

(b) In Area A, Drawing 1521140: Schematic diagram.

(c) The index mark, as indicated on Drawing 1521140.

NOTE

The date and place of manufacture shall not be shown.

3.3.8 Threaded parts. Bolts, nuts, screws, studs, etc., shall conform to the requirements of MIL-B-857 or FF-S-86, FF-S-92, and FF-N-836. Screw threads for all threaded parts shall be in accordance with Screw Thread Standards for Federal Services, Handbook H28, and shall be of size 2-56, 4-40, 6-32, 8-32, 10-32, 1/4-20, or 1/4-28. If larger sizes are needed, they shall be of the course-thread series. Threaded parts (other than bolts, screws, studs, nuts, etc.) requiring threads of special diameter and pitch shall employ the special pitch of 16 threads to the inch. For applications where a finer pitch is necessary, one of the following shall be selected: 20, 28, 36, 44, or 56.

3.3.8.1 Thread projection. Screws or bolts secured by nuts or other retaining devices shall be of such length that a minimum of 1-1/2 threads and a maximum of 1/8 inch plus 1-1/2 threads for screws up to one inch in length and 1/4 inch plus 1-1/2 threads for screws over one inch in length, shall project beyond the retainer. These requirements shall not be construed to preclude the use of screws assembled in blind holes, spacers, etc., or where design restrictions require that the threaded portion of the screw be flush with the retaining device. In no case shall the screw be clipped, bent, or otherwise shortened after plating.

3.3.8.2 Thread engagement. Screws and bolts in tapped parts other than nuts shall enter and engage for a distance at least equal to the diameter of the screw or bolt.

3.3.8.3 Setscrews. Where setscrews are used, they shall be in accordance with the requirements of Type G or H of FF-S-86.

3.3.8.4 Self-tapping screws. Self-tapping screws will not be acceptable except where employed in approved components and where their removal from the component is a remote possibility.

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3.3.8.5 Locking devices. Locking devices shall be provided for all screw-thread assemblies except for captive screws on withdrawable units. The working loose of threaded parts during tests or in service will be considered a defect in design. All lockwashers shall conform to the standards of Section 3 of OSTD 600 or applicable Military Standards and shall be fabricated of corrosion-resistant steel, plated phosphor-bronze, or steel zinc-plated in accordance with QQ-Z-325, Type II, Class 2 or 3.

3.3.8.5.1 External-tooth lockwashers shall not be used with screws 1/4 inch or greater in diameter.

3.3.8.5.2 Spring-type lockwashers may be used for screws of all sizes.

3.3.9 Methods of wiring

3.3.9.1 No more than three wires, including wires from parts, shall terminate at any one terminal.

3.3.9.2 Pressure contacts. Electrical connections shall not depend upon wires, lugs, terminals, and the like clamped between a metallic and an insulating or pliable material. Connections of this type shall be clamped between rigid metallic members, such as in an assembly of a machine screw, two nuts, and two washers. If such an assembly cannot be used and maintenance of a tight connection depends upon the resistance of an insulating or pliable material to compressive stress or shear, the connection shall be securely soldered.

3.4 Construction and design

3.4.1 General

3.4.1.1 Objective. The basic design objectives are that the equipment will meet the needs of the Naval service and that the final product will reflect the utmost in simplicity, have maximum reliability consistent with the state of the art, be easy to operate and maintain, and be economical to produce.

3.4.1.2 Needs of the Naval service. The equipment shall be designed to withstand storage and service conditions over long periods.

3.4.1.3 Reliability. It is imperative that reliability of operation be considered of prime importance.

3.4.1.4 Simplicity. The equipment shall represent the simplest design consistent with the operational and environmental conditions set forth in this specification.

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3.4.2 Dimensions and tolerances. Dimensions and tolerances of the relay shall conform to those shown on Drawing 1521140.

3.4.3 Weight. The weight of the relay shall not exceed three ounces.

3.4.4 Mechanical design

3.4.4.1 Type. The relay shall be the self-aligning type, and shall remain in adjustment in any position.

3.4.4.2 Coil mounting. The coil of the relay shall be mounted in such a manner that it will not loosen or become displaced by any tests of environmental conditions described herein.

3.4.4.3 Adjustments. All adjustable features of the relay shall be so designed that the setting of any adjustment will not be altered by any tests or environmental condition described herein. There shall be no external adjustments.

3.4.4.4 Enclosure. The relay shall be sealed. The term "sealed" is construed to cover all methods of sealing; namely, vacuum or atmospheric, and pressurized with dry air or inert gas. When tested as specified in 4.5.4.3 the sealed relay shall show no evidence of leakage.

3.4.4.5 Mounting. The relay shall be mounted by two number 4-40 studs, located as shown on Drawing 1521140.

3.4.5 Electrical design

3.4.5.1 Contacts. The contacts of the relay shall be of the proper size, material, and type to carry, to make, and to break the rated current. The contacts shall be securely fastened to the stationary and movable current-carrying members.

3.4.5.1.1 Contact rating. The rated current of the contacts shall be 2.0 amperes at 28 volts resistive load.

3.4.5.2 Contact arrangement. The contact arrangement of the relay shall be double-pole double-throw, as shown in Drawing 1521140. With the relay coil deenergized, contact 4 shall be closed to contact 3, and contact 8 shall be closed to contact 7. The schematic diagram of the relay shown on the above drawing shall be included in the marking of the relay (3.3.7.1) in Area A.

3.4.5.3 Coil. The coil of the relay shall be evenly wound and properly insulated from the frame and other grounded parts.

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3.4.5.4 Case. The case of the relay shall not be used for grounding nor shall it be electrically connected to any terminal of the relay.

3.4.5.5 Terminals. Eight J-hook type terminals, each capable of accommodating two number 18 wires, shall be provided and located as shown on Drawing 1521140. Terminals shall be hot-tin dipped or solder tinned.

3.4.6 Welding. All welds shall be of ample size with thorough penetration and good fusion, and shall be substantially free of injurious scabs, blisters, abnormal pockmarks, bruises, or other surface defects. Welds shall be capable of withstanding a strength test to destruction in which failure shall not occur within the weld itself. There shall be no evidence of harmful effects due to the welding process on the material adjacent to the weld. All welds shall be cleaned of scale, oxidation products, and excess flux by sand blasting, wire brushing, buffing, or other suitable means. Surfaces cleaned with acid shall be completely neutralized. Where tack or spot welds are employed, the number of welds shall be sufficient to provide adequate strength; at least two welds shall be used to hold any part.

3.4.7 Soldering. Only class SN60 solder in accordance with QQ-S-571 shall be used for electrical connections. Soldering processes shall conform to the requirements specified in MIL-S-6872. No assembly shall depend solely on soft solder for mechanical strength except for relatively light parts which have by actual use proved to be suitable.

3.4.8 Brazing. All brazed joints shall be in accordance with MIL-B-7883, and shall be of ample size to meet the individual requirements of the parts concerned.

3.4.9 Riveting. Riveting operations shall be carefully performed in order to ensure that the rivets are tight and satisfactorily headed.

3.5 Performance requirements and product characteristics. The relay shall meet the following performance requirements and product characteristics:

3.5.1 Dielectric strength. The relay shall withstand the application of an ac voltage of 700 volts rms at 60 Hertz (Hz) for a period of one minute between any relay terminal and the relay case, and between any coil terminal and any contact terminal without damage, arcing, or breakdown.

3.5.2 Insulation resistance. The insulation resistance between any terminal of the relay and the case, and between any coil terminal and any contact terminal, shall be not less than 100 megohms.

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3.5.3 Contact resistance. The resistance across contacts 3 and 4 are contacts 7 and 8 when the relay coil is deenergized, and across contacts 2 and 4 and across contacts 6 and 8 when the coil is energized and the contacts are carrying rated current (3.4.5.1.1), shall not exceed 0.05 ohm.

3.5.4 Contact bounce. The relay shall not exhibit any characteristics of contact bounce or chatter three milliseconds after the contacts make or break.

3.5.5 Coil resistance. The coil resistance of the relay shall be 1000 ohms plus or minus 10 percent, when measured at a temperature of 68 degrees F (20 degrees C).

3.5.6 Coil current

3.5.6.1 Operate. The maximum current at which contact 4 transfers from contact 3 to contact 2 and contact 8 transfers from contact 7 to contact 6 shall be not greater than 6.3 milliamperes. Under the conditions of vibration (3.6.1.2) and acceleration (3.6.3) contact 4 shall remain closed to contact 3 and contact 8 shall remain closed to contact 7 with a coil current increasing up to, but not exceeding, 5.0 milliamperes.

3.5.6.2 Release. The minimum current at which contact 4 transfers from contact 2 to contact 3 and contact 8 transfers from contact 6 to contact 7 shall be not less than 2.3 milliamperes. Under the conditions of vibration (3.6.1.2) and acceleration (3.6.3) contact 4 shall remain closed to contact 2 and contact 8 shall remain closed to contact 6 with a coil current decreasing to, but not below, 3.15 milliamperes.

3.5.7 Sensitivity

3.5.7.1 Operate. The operating sensitivity shall be a nominal 40 milliwatts.

3.5.7.2 Release. The release sensitivity shall be a nominal 5 milliwatts.

3.5.8 Operating and releasing time

3.5.8.1 Operating. The operating time, when the coil of the relay is energized from a nominal 10 volt dc source, shall be not greater than 30 milliseconds.

3.5.8.2 Releasing. The release time, when an energized coil is deenergized, shall be not greater than 10 milliseconds.

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3.5.9 Operating life. The relay shall be capable of meeting the performance and product characteristics specified herein after experiencing 100,000 contact transfers under the environmental conditions described in 4.4.1.

3.6 Environmental requirements. The relay shall meet the other applicable requirements of this specification during or after the tests defined or specified in 4.5.

3.6.1 Vibration. The relay shall meet the following vibration requirements:

3.6.1.1 Vibration, nonoperating. The relay shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.5.1.2.

3.6.1.2 Vibration, operating. The relay shall show no damage and shall meet all the performance requirements and product characteristics of this specification during and after the test specified in 4.5.1.3.

3.6.1.3 Vibration, transportation. The relay shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.5.1.4.

3.6.2 Shock, transportation. The relay shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.5.2.

3.6.3 Temperature and humidity

3.6.3.1 High temperature and humidity, nonoperating. The relay shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.5.3.1.

3.6.3.2 High temperature, operating. The relay shall be capable of meeting all the performance requirements of this specification while undergoing the test specified in 4.5.3.2.

3.6.3.3 Low temperature, nonoperating. The relay shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.5.3.3.

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3.6.3.4 Low temperature, operating. The relay shall be capable of meeting all the performance requirements of this specification while undergoing the test specified in 4.5.3.4.

3.6.4 Pressure

3.6.4.1 High altitude transportation. The relay shall withstand exposure to a pressure 1.7 pounds per-square-inch absolute (psia) as specified in 4.5.4.1 and thereafter shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

3.6.4.2 Fluctuating pressure. The relay shall withstand exposure to pressure cycles between atmospheric pressure and 10 psia as specified in 4.5.4.2 and thereafter shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

3.6.4.3 Immersion. The relay shall withstand exposure to immersion as specified in 4.5.4.3 and thereafter shall show no damage or corrosion and shall meet all the performance requirements and product characteristics of this specification.

3.7 Cleanup. Prior to and after final assembly all parts components, and the assembly shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, rosin flash, and other foreign matter.

3.8 Workmanship. The relay, including all parts and accessories, shall be constructed and finished in a manner to ensure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, painting, plating, riveting, machine screw assemblies, welding, brazing, and freedom of parts from burrs and sharp edges. The standards of workmanship exhibited in any approved preproduction sample subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection of the relays shall be classified as follows:

(a) Preproduction inspection (paragraph 4.5)

(b) Quality conformance inspection (paragraph 4.4)

(c) Periodic production inspection (paragraph 4.5)

4.3 Sampling

4.3.1 Preproduction. A preproduction sample of five relays, manufactured in accordance with 3.3.3, shall be subjected to the preproduction tests detailed in 4.5 at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the relays by the supplier, prior to the approval of the preproduction sample, shall be at the supplier's risk.

4.3.2 Lot size. Unless otherwise specified in the contract or order, each shipment of units of product shall be a separate "inspection lot" and may differ from the quantity specified in the total order or contract. In any event, all units of a single "inspection lot" shall be made to the same design, with the same materials and manufacturing techniques, and shall be part of one continuous production run.

4.3.3 Periodic production samples. From each succeeding lot or grand lot of 300 or less that have passed the quality conformance inspection of 4.4, a random sample of five relays will be selected for submission to an activity designated by the procuring activity to determine compliance of the samples with the requirements of the contract, specification, and drawings. The relays shall be considered satisfactory if they meet all specification and pertinent drawings requirements and withstand the tests specified in 4.4 and 4.5. Failure of the periodic production sample to comply with these requirements will result in the rejection of the lot, as determined by the procuring activity.

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4.4 Quality conformance inspection. Each relay delivered under contract shall be subjected to the following tests and examinations. Failure to meet any requirement specified herein shall be considered cause for rejection.

4.4.1 Test conditions. Unless otherwise specified, the relay shall be subjected to quality conformance tests under the following conditions:

4.4.1.1 Temperature. Room ambient 65 degrees F to 95 degrees F (18 degrees C to 35 degrees C).

4.4.1.2 Altitude. Normal ground.

4.4.1.3 Vibration. None.

4.4.1.4 Humidity. Room ambient to 95 percent relative maximum.

4.4.2 Test and inspection equipment and facilities. The manufacturer shall furnish and maintain all necessary test equipment and facilities and shall provide personnel for performing all quality conformance tests. The test equipment shall be adequate in quantity and, unless definite requirements are specified, shall be of sufficient accuracy and quality to permit the performance of the required test .

4.4.3 Items of test equipment. The following items of test equipment are required to perform the quality conformance tests set forth in this specification:

4.4.3.1 A source of 700 volt, rms, 60 Hertz, ac power.

4.4.3.2 A 500 volt dc megohmmeter, with a midscale reading of not less than 100 megohms (or a megohm bridge).

4.4.3.3 A Kelvin bridge, or equivalent, having a nominal 35-volt source of dc power, a resistance rated at 1-1/2 watts and variable from 5000 to 35,000 ohms, a 1-1/2 ohm resistance rated at 70 watts, and a voltmeter, scale 0 to 50 volts, accurate to within plus or minus 1.0 percent.

4.4.3.4 An oscillograph of a make satisfactory to the Naval Ordnance Systems Command or its appointed agent. An interrupter (or signal generator) capable of 6 Hz, with an output of 6 volts. A dc milliammeter, scale 0 to 10 milliamperes, accurate to within plus or minus 1 percent. Three dc ammeters, scale 0 to 5 amperes, accurate to within plus or minus 1 percent.

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4.4.4 Test procedure

4.4.4.1 Dielectric strength. Apply ac power source 4.4.3.1 across each of contact terminals 2, 3, 4, 6, 7, and 8 and the relay case for a period of one minute plus or minus five seconds. Apply this power source across coil terminal 1 (or 5) and each of contact terminals 4 and 8 for a like period. The relay shall meet the requirements specified in 3.5.1.

4.4.4.2 Insulation resistance. With the relay deenergized, measure the insulation resistance between the following points, using the megohmmeter or bridge of 4.4.3.2:

(a) Terminals 1, 2, 3, 6, and 7 and the relay case.

(b) Terminals 2, 3, 6, and 7 and terminal 1 (or 5).

The relay shall meet the requirements of 3.5.2.

4.4.4.3 Contact resistance. With the relay deenergized, measure the resistance across contacts 3 and 4 and across contacts 7 and 8, using the Kelvin bridge or equivalent. With the rated current of 2 amperes flowing through contacts 3 and 4 and through contacts 7 and 8, measure the resistance across each pair of contacts. With the relay coil energized to close contact 4 to contact 2 and contact 8 to contact 6, and with the rated current flowing through these contacts, measure the resistance across each pair of contacts. The contact resistance of the contacts shall be as specified in 3.5.3. Upon completion of the operating life test (4.5.6), the contact resistances measured as above shall be no more than double the value specified in 3.5.3.

4.4.4.4 Contact bounce. With oscillograph 4.4.3.4, test the relay to determine whether or not it exhibits any evidence of chatter or contact bounce. The relay shall meet the requirements of 3.5.4.

4.4.4.5 Coil resistance. Measure the coil resistance of the relay by any suitable means. The coil resistance shall lie within the limits specified in 3.5.5 when corrected to the temperature indicated.

4.4.4.6 Coil current. With the relay deenergized, check continuity between terminals 3 and 4 and between terminals 7 and 8, using the dc voltage source and voltmeter 4.4.3.3. With the coil energized with a coil current of 6.3 milliamperes, check continuity between terminals 2 and 4 and between terminals 6 and 8.

4.4.4.6.1 Operate. With rated current flowing through the appropriate contact terminals, apply a nominal ten volts dc, through the variable resistance of 4.4.3.3 at maximum value and the milliammeter, across coil terminals 1 and 5. Gradually reduce the resistance value and accurately

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determine the current value at which contacts 4 and 8 transfer from 3 to 2 and from 7 to 6 respectively. The current value at which transfer takes place shall not exceed that specified in 3.5.6.1.

4.4.4.6.2 Release. With the circuit arrangement of 4.4.4.6.1 and the relay coil energized with a current of 6.3 milliamperes, gradually increase the resistance value and accurately determine the current value at which contacts 4 and 8 transfer to contacts 3 and 7 respectively. The current value at which transfer takes place shall be not less than that specified in 3.5.6.2.

4.4.4.7 Sensitivity. From the values determined in 4.4.4.5 and 4.4.4.6, compute the operating and releasing sensitivity of the relay. Sensitivity shall be as specified in 3.5.7.

4.4.4.8 Operating and releasing time. With the oscillograph and interrupter (or signal generator) of 4.4.3.4, test the relay for chatter and contact bounce and determine the operating and releasing time of the relay at a rate of six transfers per second. The relay shall meet the requirements specified in 3.5.4 and 3.5.8.

4.4.4.9 Visual and dimensional examinations. Each relay shall be examined to determine conformance with the requirements for workmanship and dimensions.

4.4.4.10 Packaging, packing, and marking. Packaging, packing, and marking shall comply with the requirements of Section 5.

4.5 Preproduction and periodic production inspection. The preproduction and periodic production samples, after satisfactorily passing the quality conformance inspections detailed in 4.4, shall be subjected to the following tests and examinations. Certified inspection data covering the results of preproduction and periodic production inspections shall be forwarded to the procuring activity.

4.5.1 Vibration

4.5.1.1 Mounting. The relay, less pack or packaging, shall be mounted securely to a vibration device by its mounting holes and vibration tested as specified in 4.5.1.2 and 4.5.1.3.

4.5.1.2 Vibration, nonoperating. The relay shall be subjected to the "Vibration, nonoperating" test of MIL-E-82590(OS) and thereafter shall meet the requirements of 3.6.1.1.

4.5.1.3 Vibration, operating. The relay shall be subjected to the "Vibration, operating" test of MIL-E-82590(OS) and during and thereafter shall meet the requirements of 3.6.1.2.

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4.5.1.4 Vibration, transportation. The relay shall be subjected to the "Vibration, transportation" test of MIL-E-82590(OS) and thereafter shall meet the requirements of 3.6.1.3.

4.5.2 Shock, transportation. The relay shall be subjected to the "Shock, transportation" test of MIL-E-82590(OS) and thereafter shall meet the requirements of 3.6.2.

4.5.3 Temperature and humidity

4.5.3.1 High temperature and humidity, nonoperating. The relay shall be subjected to the "High temperature and humidity, nonoperating" test of MIL-E-82590(OS) and thereafter shall meet the requirements of 3.6.3.1.

4.5.3.2 High temperature, operating. The relay shall be subjected to the "High temperature, operating" test of MIL-E-82590(OS) and, while at the test temperature, shall meet the requirements of 3.6.3.2.

4.5.3.3 Low temperature, nonoperating. The relay shall be subjected to the "Low temperature, nonoperating" test of MIL-E-82590(OS) and thereafter shall meet the requirements of 3.6.3.3.

4.5.3.4 Low temperature, operating. The relay shall be subjected to the "Low temperature, operating" test of MIL-E-82590(OS) and, while at the test temperature, shall meet the requirements of 3.6.3.4.

4.5.4 Pressure

4.5.4.1 Altitude. The relay shall be subjected to pressure testing as specified in Section 4 of MIL-T-18404 for altitude. At the conclusion of this test the relay shall meet the performance and other requirements specified in this specification.

4.5.4.2 Fluctuating. The relay shall be subjected to pressure cycling as specified in Section 4 of MIL-T-18404 for a submarine launched torpedo. At the conclusion of this test the relay shall meet the performance and other requirements specified in this specification.

4.5.4.3 Immersion. The relay shall be immersed in a saturated solution of sodium chloride under an absolute pressure of 2.5 inches of mercury for a period of four hours. The immersed relay shall be observed for leakage, as evidenced by bubbles emanating from the sealed case. Approximately one-half the relays so immersed shall have the case removed in order that the interiors may be further examined for evidence of leakage.

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4.5.5 Operating life. Operate the relay, under the environmental conditions specified in 4.4.1, for a total of 100,000 contact transfers while the contacts are carrying the rated current of 2.0 amperes at 28 volts resistive load. The relay coil will be energized from a 10 plus or minus 1 volts dc or rectified ac power source. The test pulse rate shall be at least 4 Hertz (4 Hz) with a 50 plus or minus 10 percent duty cycle. At the conclusion of the test the relay shall meet the performance and other requirements specified in the specifications.

4.6 Waiver of tests. Tests specified in 4.5.1.4 and 4.5.2 above may be waived by the procuring agency for Level C packs only, provided the contractor guarantees safe delivery to the using activity.

4.7 Functional tests. At the conclusion of the tests specified in 4.5.1.4 and 4.5.2 above, functional and other tests, as applicable, shall be conducted on the relay to determine freedom from operational malfunction and compliance with all performance requirements and product characteristics of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. The levels of preservation and packaging shall be as specified (see 6.2).

5.1.1 Level A

5.1.1.1 Unit packaging. The relay shall be packaged in accordance with Method IIB of MIL-P-116.

5.1.2 Level B. Not applicable.

5.1.3 Level C

5.1.3.1 Unit packaging. The relay shall be packed in accordance with Method III of MIL-P-116.

5.2 Packing. Packing shall be Level A or Level C, as specified (see 6.2).

5.2.1 Level A

5.2.1.1 Consolidation container. The relays, packaged as described in 5.1.1.1, shall be packed in a wooden box in accordance with PPP-B-621, Class 2.

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5.2.1.2 Palletizing. When specified, Level A packs shall be palletized in accordance with MIL-STD-147, Pattern No. 95.

5.2.2 Level B. Not applicable.

5.2.3 Level C. The relays, packaged in accordance with 5.1.3.1, shall be packed to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with common carrier regulations applicable to the mode of transportation.

5.3 Levels A or C cushioning. The unit package shall incorporate sufficient cushioning material, bracing or other adequate shock absorbing devices to ensure that the relay will meet all the requirements of Section 3 after being tested in accordance with paragraph 4.5.1.4 (Vibration) and 4.5.2 (Shock). At the time of packaging or packing, the moisture content of the cushioning material shall be not more than 5 percent.

5.4 Marking

5.4.1 Special markings. None, unless otherwise specified.

5.4.2 Normal markings. In addition to the markings required by the contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129, including serial numbers.

6. NOTES

6.1 Intended use. The relay is intended for use in Torpedo Mk 37 Mod 2.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Any exceptions to this specification.

(c) Required level of preservation and packaging (see 5.1).

(d) Required level of packing, including palletizing (see 5.2).

(e) Assigned activity for preproduction and periodic production inspections.

(f) Data required other than that specified in 3.3.2.

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(g) That additional samples required because of failure of any unit(s) to pass the prescribed periodic production inspections shall be provided and inspected at the expense of the supplier.

6.3 Preparation for delivery criteria. Criteria for use of the proper level of preservation, packaging, and packing shall be as follows.

Level A. This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

Level C. This level shall be used only when it is definitely known that the packaged items are to be shipped to domestic installations for immediate use at the first receiving activity.

6.4 Inherent or latent defects. When material which has been inspected, tested, and accepted contains inherent or latent defects subsequently exposed, the contractor may be required to replace the defective material without expense to the Government.

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